

Causal Inference in Economics and Marketing

Section 05: Instrumental Variables and Unobserved Confounding

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1 When conditioning fails

1.1 Two graphs we cannot tell apart

Sections 03 and 04 assumed that, given observed controls, treatment is as good as random. Every estimator there, from regression to double machine learning, was a way to condition well. None of them helps when a variable that drives both the treatment and the outcome is not observed.

The classic example is schooling and earnings. Ability raises schooling and raises earnings. Suppose we observe schooling D , earnings Y , and background X , but not ability A . Two structural models are consistent with any joint distribution of (Y, D, X) : one in which D causes Y , and one in which A causes both and D has no effect. No amount of data separates them (Chernozhukov et al. 2026, ch. 12). Conditioning on X is not enough because A is not in X .

The same problem has other names: demand estimation (a demand shock moves both price and quantity), program evaluation with self-selection (those who expect to gain enroll), and any marketing setting where

the firm targets the customers it expects to respond.

1.2 What can be said without an instrument

In the partially linear model

$$Y = \theta D + \delta A + f_Y(X) + \epsilon_Y, \quad D = \gamma A + f_D(X) + \epsilon_D,$$

partialling out X and regressing $\tilde{Y}$ on $\tilde{D}$ gives $\theta + \phi$ with $\phi = \delta\gamma/E[\tilde{D}^2]$. The bias is bounded in terms of two partial R^2 : how much of the residual variation in $\tilde{D}$ the confounder explains ($R_{D \sim A}^2$), and how much of the residual variation in $\tilde{Y}$ it explains after $\tilde{D}$ ($R_{Y \sim A|D}^2$):

$$\phi^2 = \frac{R_{Y \sim A|D}^2 R_{D \sim A}^2}{1 - R_{D \sim A}^2} \cdot \frac{E[(\tilde{Y} - \beta \tilde{D})^2]}{E[\tilde{D}^2]}$$

(Chernozhukov et al. 2026, Theorem 12.2.1; Cinelli and Hazlett 2020; appendix 11.2). The second factor is estimated from the DML residuals. The first is the analyst’s belief about the confounder, often benchmarked against the observed controls.

`dml_sensitivity()` applies the bound to a `cm_dml` fit. The running “linear” design of this lecture has an unobserved A with $\delta = 1$ and $\gamma = 0.8$, so the partialling-out estimate is biased upward.

```
dat <- sim_iv(n = 2000, dgp = "linear", seed = 1)
rf <- iv_forest()
fit_plr <- est_dml(dat, "y", "d", x5, learner_l = rf, learner_m = rf, seed = 1)
sens <- dml_sensitivity(fit_plr, r2_y = 0.10, r2_d = 0.10)
sens
#> Sensitivity of the DML estimate to an unobserved confounder
#> estimate = 1.2724 (SE 0.0192)
#> benchmark R2_y = 0.1, R2_d = 0.1: |bias| <= 0.0849, bounds [1.1875, 1.3573]
#> robustness value (equal R2 that moves the estimate to zero): 0.765; with the 95% interval: 0.756
```

The estimate is 1.27 for a true effect of 1. A confounder that explains 10 percent of the residual variation in both variables would move it by 0.08, so the interval of estimates consistent with that scenario is [1.19, 1.36]. The robustness value says a confounder would need partial R^2 of about 0.77 on both sides to bring the estimate to zero. In this design the true confounder is that strong. Figure 1 shows the contours.

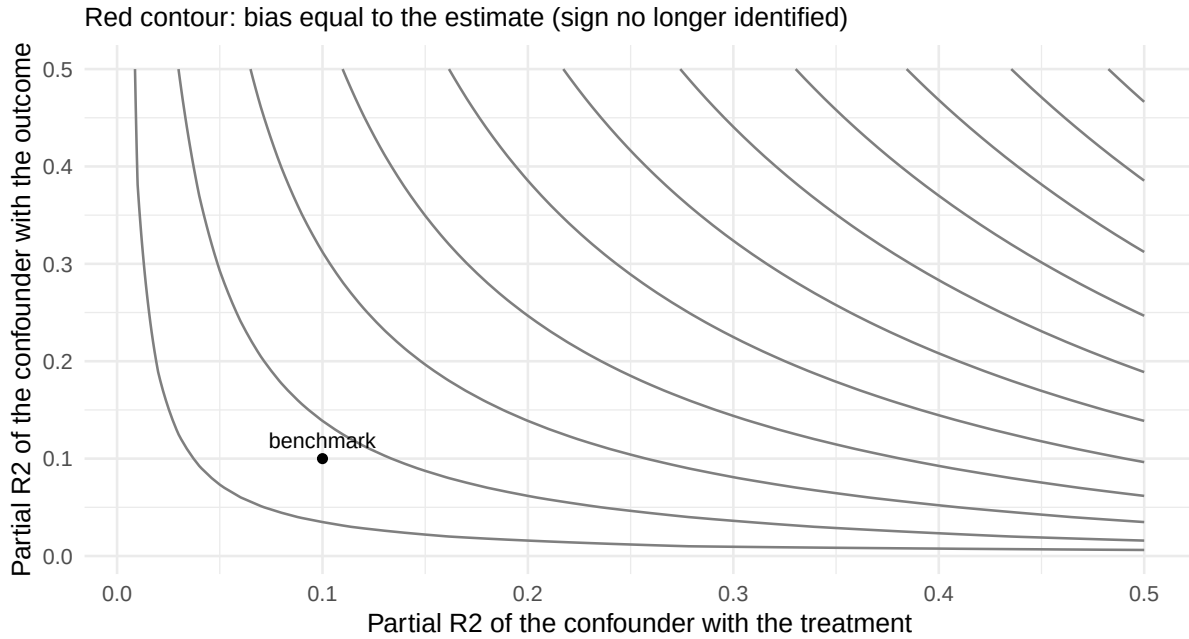


Figure 1: Contours of the omitted-confounder bias bound as a function of the two partial R-squareds. The red contour marks confounders strong enough to bring the estimate to zero.

Sensitivity analysis tells us how wrong we could be. It does not tell us the answer. For that we need a second source of variation.

1.3 Instruments and Wright's path diagram

An instrument Z is a variable that moves the treatment but affects the outcome only through the treatment. Philip Wright (1928) drew the first causal path diagram to make the point: if $Z \rightarrow D \rightarrow Y$ and Z has no other path to Y , then the effect of Z on Y equals the effect of Z on D times the effect of D on Y . Two regressions we can run give the product and one factor; the ratio is the effect we want.

Examples that appear in this lecture: an offer to participate (a nudge) as an instrument for participation; a supply shifter (weather where the product is grown) for price; distance to college for schooling; random assignment of judges for incarceration; national industry shocks apportioned by local industry shares for local exposure to trade.

The price is a new assumption. Conditional exogeneity said “no unobserved confounder of D ”. The exclusion restriction says “no unobserved path from Z to Y except through D ”. Whether that is more or less credible depends entirely on where Z comes from. When Z is randomized, it is credible by design.

1.4 Roadmap

Section 2 sets up instrumental variables in the linear model and the reporting conventions. Section 3 asks what IV estimates when effects are heterogeneous. Section 4 develops the control-function approach, which equals two-stage least squares in the linear model and extends to nonlinear ones. Section 5 gives the orthogonal scores that let machine learning handle the controls. Section 6 treats weak instruments. Section 7 covers several instruments, judges, marginal treatment effects, and constructed instruments. Section 8 turns to experiments and platforms, Section 9 to traps, Section 10 to the replications. The appendix holds the proofs.

2 Instrumental variables in the linear model

2.1 Relevance, exclusion, and the Wald ratio

Take the structural equation $Y = \alpha + \theta D + e$ with $\text{Cov}(D, e) \neq 0$. An instrument Z satisfies

- **relevance:** $\text{Cov}(Z, D) \neq 0$, and
- **exogeneity with exclusion:** $\text{Cov}(Z, e) = 0$.

Then $\text{Cov}(Z, Y) = \theta \text{Cov}(Z, D)$, so

$$\theta = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)} = \frac{E[Y | Z = 1] - E[Y | Z = 0]}{E[D | Z = 1] - E[D | Z = 0]}$$

for a binary instrument: the Wald ratio, reduced form over first stage. In an encouragement design the numerator is the intention-to-treat effect and the denominator is the compliance rate; the ratio rescales the ITT to the effect of the treatment itself.

2.2 Two-stage least squares

With controls X and one or more instruments, two-stage least squares (2SLS) regresses D on (Z, X) , forms $\hat{D}$, and regresses Y on $(\hat{D}, X)$. Equivalently, it solves the moment condition $E[(Y - \alpha - \theta D - X'\beta)(Z, X)] = 0$. Three facts to keep in mind.

- The second-stage OLS standard errors are wrong, because they use residuals computed with $\hat{D}$ rather than D . Software reports the correct ones; hand-rolled two stages do not.
- With more instruments than endogenous regressors the model is overidentified and 2SLS combines the instruments with weights from the first stage (Section 7.1).
- “Endogenous” is a statement about a structural error, not about correlation in the data. It is meaningful only once the structural equation is written down.

The running design has one instrument, five covariates, and a true effect of 1. The instrument is excluded conditional on $h(x) = 0.5x_1^2 + 0.5\sin(2x_2)$, which also enters the outcome; in this section we include the two terms as controls, as an analyst who knew the functional form would. Section 5 drops that knowledge. Every regression an applied paper reports appears in one table.

```
dat$x1sq <- dat$x1^2
dat$s2x2 <- sin(2 * dat$x2)
xc <- c("x1sq", "s2x2", "x3", "x4", "x5")
ols <- lm_robust(y ~ d + x1sq + s2x2 + x3 + x4 + x5, data = dat)
first <- lm_robust(d ~ z + x1sq + s2x2 + x3 + x4 + x5, data = dat)
reduced <- lm_robust(y ~ z + x1sq + s2x2 + x3 + x4 + x5, data = dat)
tsls <- iv_robust(y ~ d + x1sq + s2x2 + x3 + x4 + x5 | z + x1sq + s2x2 + x3 + x4 + x5, data = dat)
```

Table 1: The four regressions to report. True effect 1. The Wald ratio of the two middle rows equals the 2SLS row.

Regression	Coefficient	Estimate	SE
OLS: y on d, X	d	1.282	0.019
First stage: d on z, X	z	1.026	0.029
Reduced form: y on z, X	z	1.023	0.051
2SLS: y on d, X (z instrument)	d	0.997	0.031

OLS is biased upward by the confounder, as Section 1.2 anticipated. The 2SLS estimate is close to 1 with a standard error about 1.6 times the OLS one: the price of using only the variation the instrument induces.

2.3 Standard errors

Report heteroskedasticity-robust standard errors by default and cluster at the level at which the instrument varies (a policy that varies by state, weather that varies by market-day, a judge assigned to a court-year cell). `iv_robust()` and `feols()` do both. With `fixest`:

```
fe <- feols(y ~ x1sq + s2x2 + x3 + x4 + x5 | d ~ z, data = dat, vcov = "hetero")
coeftable(fe)["fit_d", ]
#>      Estimate      Std. Error      t value      Pr(>|t|)
#> 9.966124e-01 3.051296e-02 3.266194e+01 8.833946e-188
fitstat(fe, type = c("ivf", "ivwald"))
#> F-test (1st stage), d: stat = 1,253.5, p < 2.2e-16, on 1 and 1,993 DoF.
#> Wald (1st stage), d : stat = 1,213.4, p < 2.2e-16, on 1 and 1,993 DoF, VCOV: Heteroskedasticity-robust.
```

2.4 Overidentification and endogeneity tests

With more instruments than endogenous regressors, the Sargan-Hansen test asks whether the instruments give compatible estimates. A rejection means that at least one instrument violates exclusion, *or* that the instruments identify different effects for different people (Section 3). A non-rejection with instruments that share a bias is uninformative. Treat it as a test of the homogeneous-effect functional form rather than as a license.

The Wu-Hausman test compares OLS and 2SLS. It tests whether endogeneity matters, not whether the instrument is valid, and has low power when the instrument is weak (2SLS then drifts toward OLS). Section 4 shows it is the t test on the control function.

2.5 Reporting

A first stage, a reduced form, and the IV estimate, always together, with the same controls and the same sample. The reader needs the first stage to judge relevance and the reduced form to see the raw effect of the instrument; a paper whose abstract does not say what the instrument is and why it is excluded has usually not thought hard about either.

Wayne’s earlier note on Oreopoulos (2006) is the template: a change in the minimum school-leaving age in Britain (1947) and Northern Ireland (1957) instruments years of schooling in a regression of log earnings, with a quartic in birth cohort as controls. Its Table 1 reports the first stage (the jump in years of schooling at the reform cohort), the reduced form (the jump in earnings), and the IV ratio side by side, for each country and pooled with a country fixed effect. That table structure is what to imitate.

3 Heterogeneous effects: what IV estimates

3.1 Potential outcomes indexed by the instrument

Let $D(z)$ be the treatment a unit would take under instrument value z and $Y(d)$ the outcome under treatment d . With a binary instrument and a binary treatment, each unit is one of four types (Imbens and Angrist 1994):

$D(0)$	$D(1)$	Type
0	1	complier
1	1	always-taker
0	0	never-taker
1	0	defier

The LATE assumptions are: (i) independence, $Z \perp (Y(0), Y(1), D(0), D(1))$ (conditionally on X if needed); (ii) exclusion, Y depends on Z only through D ; (iii) relevance, $P(D(1) = 1) \neq P(D(0) = 1)$; (iv) monotonicity, no defiers. Monotonicity is natural when the instrument is an offer of something people want, and much less

natural when the instrument is an assigned decision maker, since two judges need not order every defendant the same way (Section 7.2).

3.2 The LATE theorem

Under (i)-(iv),

$$\frac{E[Y | Z = 1] - E[Y | Z = 0]}{E[D | Z = 1] - E[D | Z = 0]} = E[Y(1) - Y(0) | D(1) > D(0)],$$

the average effect among compliers. *Skeleton.* By independence and exclusion the numerator is $E[Y(D(1)) - Y(D(0))] = E[(Y(1) - Y(0))(D(1) - D(0))]$. The difference $D(1) - D(0)$ is 1 for compliers, 0 for always- and never-takers, and -1 for defiers; without defiers the numerator is the complier effect times the complier share. The denominator is the complier share. Divide (appendix 11.1).

Three consequences.

- The IV estimand is not the ATE. It is the effect for the units the instrument moves. Different instruments move different people and estimate different LATEs; overidentification tests can reject for that reason alone.
- The complier share is estimable (the first stage), and so are the compliers' characteristics (Section 3.4), even though no individual complier can be named.
- With a continuous or multi-valued instrument, IV is a weighted average of effects for the units moved at each margin (Sections 7.1 and 7.3).

3.3 IV as a compliance-weighted average

With heterogeneous effects β_i and individual compliance $\pi_i = P(D_i = 1 | Z_i = 1) - P(D_i = 1 | Z_i = 0)$, the Wald ratio converges to $E[\pi_i \beta_i] / E[\pi_i]$: units the instrument moves more get more weight. It equals the ATE if effects are homogeneous or if β_i is uncorrelated with π_i . When treatment is only available through the instrument ($\pi_i = p_{1i}$), the units most likely to take it up when offered carry the weight; if those are the units with the largest gains, IV exceeds the ATE. When treatment is compulsory under $Z = 1$ and voluntary otherwise ($\pi_i = 1 - p_{0i}$), the sign reverses.

The running “late” design makes this concrete: compliers have effect $2 + x_1$, always-takers $3 + x_1$, never-takers $0.5 + x_1$. The population ATE and the LATE differ.

```
d1 <- sim_iv(n = 4000, dgp = "late", seed = 2)
c(late = attr(d1, "late"), ate = attr(d1, "ate"), complier_share = attr(d1, "complier_share"))
#>      late      ate complier_share
#>  2.045009  1.931988  0.605000
wald <- iv_robust(y ~ d | z, data = d1)
c(wald = coef(wald)[["d"]], se = wald$std.error[["d"]])
#>      wald      se
#> 3.2310542 0.1208871
```

The Wald estimate is the LATE, not the ATE, and nothing in the data would reveal the always-takers' larger effect.

3.4 Who the compliers are

Abadie (2003) shows that the weights

$$\kappa_i = 1 - \frac{D_i(1 - Z_i)}{1 - \pi(X_i)} - \frac{(1 - D_i)Z_i}{\pi(X_i)}, \quad \pi(X) = P(Z = 1 | X),$$

average to the complier share and that $E[\kappa g(X)] / E[\kappa]$ is the complier mean of any function of X . The intuition: units with $D = 1, Z = 0$ are always-takers and units with $D = 0, Z = 1$ are never-takers; subtracting them (with inverse-propensity weights that undo the assignment of Z) leaves compliers. The same logic gives the always-taker and never-taker means. `complier_profile()` reports the table.

```

cp <- complier_profile(dl, d = "d", z = "z", covariates = c("x1", "x2"), x = "x1", n_boot = 199, seed = 1)
cp
#> Compliance groups (Abadie's kappa), n = 4000
#>      group share std.error
#> complier 0.609  0.01329
#> always_taker 0.188  0.00865
#> never_taker 0.203  0.00900
#> Covariate means by group:
#> covariate      all compliers always_takers never_takers ratio_compliers
#>      x1  0.04324  0.0390  0.102  0.00133  0.902
#>      x2 -0.00475 -0.0653  0.140  0.04343  13.736
#> se_compliers
#>      0.0277
#>      0.0274

```

The complier means of x_1 and x_2 are 0.04 and -0.07; the true values in the simulation are 0.05 and -0.05. This is the table to report whenever a LATE is the headline: it tells the reader for whom the effect was estimated, which is the first step in arguing whether it extrapolates.

3.5 Conditional LATE with covariates

When independence holds only conditionally on X , the conditional Wald ratio identifies the conditional LATE at each x , and the unconditional LATE is the ratio of averages (Frölich 2007; Chernozhukov et al. 2026, Theorem 12.4.1):

$$\theta = \frac{E\{E[Y | Z = 1, X] - E[Y | Z = 0, X]\}}{E\{E[D | Z = 1, X] - E[D | Z = 0, X]\}},$$

the ATE of Z on Y over the ATE of Z on D . Two-stage least squares with linear controls is *not* this estimand: it is the linear projection version, which coincides only when both the reduced form and the first stage are linear and additively separable in X and the compliance does not vary with X . Section 5.3 estimates the ratio of averages with doubly robust scores for each ATE and shows the difference in a Monte Carlo.

3.6 Testing the assumptions and bounding the ATE

Independence and monotonicity together imply inequalities on the observed distribution (Kitagawa 2015; Mourifié and Wan 2017): for any set A , $P(Y \in A, D = 1 | Z = 1) \geq P(Y \in A, D = 1 | Z = 0)$ and $P(Y \in A, D = 0 | Z = 0) \geq P(Y \in A, D = 0 | Z = 1)$. A violation refutes the model; the absence of a violation does not confirm it. Report the test when the instrument is not randomized.

The LATE assumptions point-identify only the complier effect. For a binary outcome, independence alone bounds the ATE for everyone: the instrument gives two sets of no-assumption bounds on $E[Y(1)]$ and $E[Y(0)]$, one per instrument value, and their intersection is valid (Manski 1990; Balke and Pearl 1997 give the sharp version). `iv_ate_bounds()` computes the intersection bounds.

```

dl$y_binary <- as.integer(dl$y > 1)
iv_ate_bounds(dl, "y_binary", "d", "z", n_boot = 199, seed = 1)
#> Manski instrumental-variable bounds (binary Y, D, Z), n = 4000
#> quantity lower upper se_lower se_upper
#> E[Y(1)] 0.693 0.894 0.01027 0.00731
#> E[Y(0)] 0.306 0.491 0.00965 0.00980
#> ATE 0.202 0.589 0.01427 0.01220
#> ATE bounds without the instrument: [-0.268, 0.732]; Wald LATE = 0.537

```

The bounds without the instrument have width one; the instrument narrows them without any monotonicity assumption. When the policy question is “treat everyone”, this is the honest answer, and the LATE is a point inside it.

4 Control functions

4.1 The idea

Two-stage least squares removes endogeneity by projection: it keeps only the part of D that the instrument predicts. A control function removes it by conditioning: it adds to the outcome equation the part of D the instrument does *not* predict. Write the first stage as $D = g(Z, X) + V$. If the dependence between D and the structural error U runs entirely through V ,

$$E[U \mid D, Z, X] = E[U \mid V],$$

then conditioning on V makes D exogenous. V is not observed, but its estimate, the first-stage residual $\hat{V}$, is. Put $\hat{V}$ in the second stage as an extra regressor. That regressor is the control function (Heckman and Robb 1985; Wooldridge 2015).

The condition is stronger than what 2SLS needs. 2SLS needs $E[UZ] = 0$; the control function needs V independent of (Z, X) , or at least $E[U \mid V, Z, X] = E[U \mid V]$ with a known (usually linear) form. In exchange it identifies the structural function and works in models where “plug in the fitted value” has no meaning.

4.2 Linear model: the same number as 2SLS

In the linear model, OLS of Y on $(D, X, \hat{V})$ gives exactly the 2SLS coefficient on D , just- or over-identified. *Skeleton.* $D = \hat{D} + \hat{V}$ where $\hat{D}$ is the first-stage fitted value. Including $\hat{V}$ as a regressor is the same as including $\hat{D}$ and $\hat{V}$ (a reparameterization). By Frisch-Waugh-Lovell, the coefficient on D in that regression is the coefficient on $\hat{D}$ after partialling out $\hat{V}$ and X ; $\hat{D}$ is orthogonal to $\hat{V}$ by construction, so this is the regression of Y on $\hat{D}$ and X , which is 2SLS (appendix 11.3).

```
cf <- cf_residuals(dat, d = "d", z = "z", x = xc)
cf_ols <- lm_robust(y ~ d + x1sq + s2x2 + x3 + x4 + x5 + v_hat, data = cf)
c(control_function = coef(cf_ols)[["d"]], tsls = coef(tsls)[["d"]])
#> control_function      tsls
#>      0.9966124      0.9966124
tidy(cf_ols)[tidy(cf_ols)$term == "v_hat", c("estimate", "std.error", "statistic", "p.value")]
#>      estimate std.error statistic      p.value
#> 8 0.4642298 0.03489606 13.30322 9.474013e-39
```

The coefficient on $\hat{V}$ is the endogeneity parameter, and its t test is the Wu-Hausman test of Section 2.4. The standard error on D printed by the second-stage regression is *not* the 2SLS standard error: $\hat{V}$ is a generated regressor, and the naive formula ignores its sampling variation. It is correct under the null of no endogeneity, which is why the t test on $\hat{V}$ is fine, and wrong otherwise. Use the 2SLS standard errors, or bootstrap both stages (Section 4.7).

4.3 Nonlinear outcomes

Now let the outcome be binary, $Y = \mathbb{1}\{\alpha + \theta D + X'\beta + U > 0\}$, with U correlated with D . Two temptations and one fix.

- **Naive probit** of Y on (D, X) ignores the endogeneity.
- **The forbidden regression** runs the probit on $(\hat{D}, X)$. It is wrong because $E[\Phi(\alpha + \theta \hat{D} + \dots) \mid Z, X] \neq \Phi(\alpha + \theta \hat{D} + \dots)$: the nonlinearity does not pass through the expectation.
- **The control function** (Rivers and Vuong 1988) runs the probit on $(D, X, \hat{V})$. Under joint normality of (U, V) , $U = \rho V + \varepsilon$ with ε independent of everything, so conditioning on V restores a probit with a rescaled index.

The coefficients of the control-function probit are scaled by $\sigma_{U|V}$, so they are not comparable to the naive ones. The object to report is the average partial effect (APE), computed from the average structural function $ASF(d, x) = E_V[\Phi(\alpha + \theta d + x'\beta + \rho V)]$, averaging over the empirical distribution of $\hat{V}$ (Blundell and Powell 2004). `cf_ape()` does the averaging; Poisson, tobit, and fractional responses use the same recipe.

Table 3: Average partial effect of an endogenous continuous regressor in a probit model, 200 simulations, $n = 2000$, error correlation 0.6. The truth is the average derivative of the structural probability.

Estimator	Bias	SD	RMSE
Naive probit	0.0786	0.0057	0.0787
Forbidden regression	0.0008	0.0096	0.0099
Control function	0.0009	0.0093	0.0089

The naive probit's APE is biased by 0.079 on a truth of about 0.15, half again too large. The forbidden regression has bias 0.001 here, which is small because the design is close to linear in the index; it is not a consistent estimator and can be far off when the nonlinearity matters. The control function has bias 0.001.

4.4 Binary endogenous regressor

When D is binary, a linear first stage is not a structural equation and its residual is not the right control: $D - \hat{p}$ is a function of the same threshold-crossing error for two different reasons. Three options.

- A **bivariate probit** models (D, Y) jointly with normal errors. It identifies the ATE, but the normality does all the work.
- A **generalized residual** from a probit first stage, $D\lambda(z'\hat{\gamma}) - (1 - D)\lambda(-z'\hat{\gamma})$ with λ the inverse Mills ratio, is the control function under the threshold-crossing model (Wooldridge 2015, Section 5); `cf_residuals(family = "probit")` returns it.
- **Give up the ATE** and report the LATE (Section 3) or the marginal treatment effect curve (Section 7.3), which is the control-function method with the propensity score as the control and no functional-form assumption on how U depends on it.

```
dp <- sim_iv(3000, dgp = "probit_cf", rho = 0.6, seed = 4)
dp$d_binary <- as.integer(dp$d > 0)
gr <- cf_residuals(dp, d = "d_binary", z = "z", x = "x1", family = "probit")
summary(gr$v_hat)
#>      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.
#> -2.370e+00 -5.119e-01 -2.131e-02  1.000e-08  4.917e-01  2.294e+00
```

4.5 Heterogeneous effects

With a random coefficient, $Y = \alpha + \beta_i D + X'\gamma + U$, the control function needs an extra term. If $E[\beta_i | V] = \bar{\beta} + \psi V$, then $E[Y | D, X, V] = \alpha + \bar{\beta} D + \psi DV + X'\gamma + \rho V$: include $\hat{V}$ and $D\hat{V}$ (Garen 1984; the correlated random coefficient model). The coefficient on D is the average effect. Heckman's (1979) selection correction is the same idea for a binary selection equation: the inverse Mills ratio is the control function. And the marginal treatment effect of Section 7.3 is what remains when the dependence of U on the control is left unrestricted.

```
cf_g <- lm_robust(y ~ d + x1sq + s2x2 + x3 + x4 + x5 + v_hat + d:v_hat, data = cf)
tidy(cf_g)[tidy(cf_g)$term %in% c("d", "v_hat", "d:v_hat"), c("term", "estimate", "std.error")]
#>      term      estimate std.error
#> 2      d  0.996543658  0.027796882
#> 8    v_hat  0.465255865  0.035986479
#> 9 d:v_hat -0.001104448  0.008498282
```

In the running design the effect is homogeneous, so the interaction is zero within sampling error and the coefficient on D does not move.

4.6 Dynamics and demand

Two applications from the replications show the reach of the method.

- **Accumulated exposure.** Hsieh et al. (2026) estimate the effect of television advertising on purchases with an ad stock $AS_{it} = \sum_l \lambda^l A_{i,t-l}$. Their instrument is realized minus expected exposure (Section

8.2); the first-stage residuals are accumulated into a control-function stock $CFS_{it} = \sum_l \lambda_{CF}^l \hat{\varepsilon}_{i,t-l}$ that enters the purchase model. The endogenous variable accumulates, so the control does too.

- **Price in discrete choice.** Petrin and Train (2010) put the residual of a regression of price on cost shifters into the utility of a multinomial logit. The vignette `usage_iv` simulates markets with an unobserved quality shock that raises both price and demand and shows the naive logit halving the price coefficient while the control function recovers it. This is the workhorse for price endogeneity in marketing when instruments exist at the product-market level but the demand model is nonlinear.

4.7 Inference for two-step estimators

Every control-function estimator has a generated regressor. Second-stage standard errors that treat $\hat{V}$ as data understate uncertainty unless the endogeneity parameter is zero. Two fixes: the Murphy-Topel correction, which needs the two scores, or a bootstrap of both stages, which needs nothing. `cf_bootstrap()` re-runs a user-supplied first and second stage on pairs (or clusters) of observations and reports standard errors and percentile intervals for coefficients and APEs.

```
first_stage <- function(df) cf_residuals(df, d = "d", z = "z", x = "x1")
second_stage <- function(df) glm(y ~ d + x1 + v_hat, data = df, family = binomial("probit"))
cf_bootstrap(dp, first_stage, second_stage, n_boot = 199, seed = 1,
  ape = function(fit, df) cf_ape(fit, df, d = "d"))
#> Two-step control-function estimator, 199 bootstrap replications
#>      term estimate std.error conf.low conf.high component
#> (Intercept) -0.4082  0.036284 -0.4824  -0.3489 coefficient
#>           d    0.9552  0.052735  0.8583   1.0595 coefficient
#>          x1    0.5679  0.043028  0.4843   0.6486 coefficient
#>         v_hat  0.6399  0.058196  0.5288   0.7434 coefficient
#>         ape_d  0.1625  0.007778  0.1464   0.1795         ape
c(true_ape = attr(dp, "ape"))
#> true_ape
#> 0.1484159
```

Cluster the bootstrap at the level at which the instrument varies. When the first stage is estimated at a coarser level than the second (prices at the product-market level, choices at the consumer level), resample at the coarser level, as the vignette’s demand example does.

5 Orthogonal scores for IV

5.1 Partially linear IV

Return to the partially linear model with an instrument,

$$Y = \theta D + f_Y(X) + U, \quad E[UZ | X] = 0,$$

where the controls X enter through an unknown function. Partial out X from everything: $\tilde{Y} = Y - E[Y | X]$, $\tilde{D} = D - E[D | X]$, $\tilde{Z} = Z - E[Z | X]$. Then $E[(\tilde{Y} - \theta \tilde{D})\tilde{Z}] = 0$ and

$$\theta = \frac{E[\tilde{Y}\tilde{Z}]}{E[\tilde{D}\tilde{Z}]},$$

Wright’s ratio on residualized variables (Chernozhukov et al. 2026, Theorem 12.3.1). The moment $\psi(W; \theta, \eta) = (Y - \ell(X) - \theta(D - r(X)))(Z - m(X))$ with nuisances $\eta = (\ell, r, m)$ is Neyman orthogonal: its derivative with respect to each nuisance vanishes at the truth (appendix 11.4). So the recipe of Section 04 applies: cross-fit the three regressions with any learner, form the residuals, and run IV on them. Standard errors are those of IV on the residuals. The score needs all three nuisances to converge at $n^{-1/4}$; there is no pair of nuisances that protects against the failure of the third, so this is orthogonal but not doubly robust.

What does θ mean when the outcome model is nonlinear? Chernozhukov et al. (2026, Theorems 12.3.2-12.3.3) show the ratio identifies the average marginal effect $E[\partial_d Y(d)]$ if either the effect of D or the effect of Z on D is homogeneous in the unobservable. When both vary with the unobservable, the ratio is a weighted average of effects, as in Section 3.3, and the LATE machinery of Section 5.3 is the right description.

5.2 Code and a Monte Carlo

In the running design the instrument is excluded only conditional on $h(x) = 0.5x_1^2 + 0.5\sin(2x_2)$, which also enters the outcome. Linear controls leave part of h in the error, and the residualized instrument is correlated with it.

```
pliv_lin <- est_dml(dat, "y", "d", x5, z = "z", model = "pliv", seed = 1)
pliv_rf <- est_dml(dat, "y", "d", x5, z = "z", model = "pliv",
  learner_l = rf, learner_m = rf, learner_z = rf, seed = 1, weak_iv = TRUE)

pliv_rf
#> Double ML estimate (partially linear IV model, theta)
#> Score: partialling_out_IV; solution: pooled across 5 fold(s); 1 repetition(s)
#> Estimate: 1.0234
#> Std. Error: 0.0302
#> 95% CI: [0.9643, 1.0825]
#> N: 2000
#> Nuisance RMSE: l_hat = 2.6, m_hat = 1.72, z_hat = 1.04
#> First stage on residuals: F (robust) = 1139.68, effective F = 1139.68
#> Anderson-Rubin 95% set (bounded): [0.9625, 1.0812]
rbind(linear = tidy(pliv_lin)[, c("estimate", "std.error")], forests = tidy(pliv_rf)[, c("estimate", "std.error")])
#> estimate std.error
#> linear 1.286938 0.01814824
#> forests 1.023440 0.03015055
```

Table 4: Partially linear IV design with a nonlinear confounder path; true effect 1; $n = 2000$; 100 simulations.

Estimator	Bias	SD	RMSE	Mean SE	95% coverage
OLS	0.401	0.014	0.401	0.014	0.000
2SLS, linear controls	0.285	0.019	0.286	0.020	0.000
DML-IV, forests	0.035	0.036	0.050	0.032	0.760

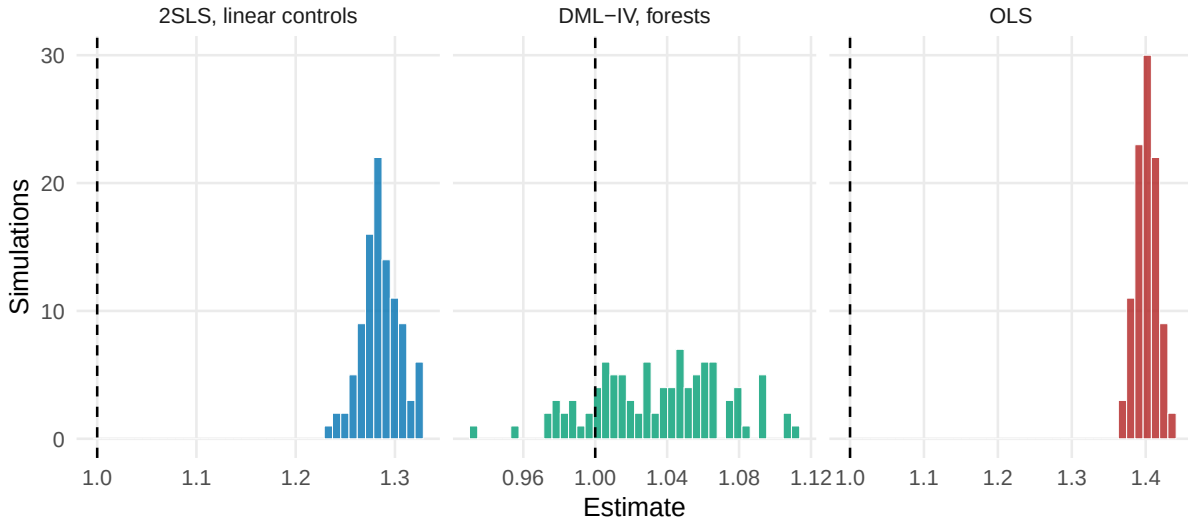


Figure 2: Sampling distributions of the three estimators in the partially linear IV design. OLS is biased by the unobserved confounder; 2SLS with linear controls is biased because the instrument is excluded only conditional on nonlinear functions of the covariates; DML-IV with forests removes most of that bias.

OLS carries the confounder bias (0.40). 2SLS with linear controls removes it and adds a different one (0.29): the instrument is not excluded given linear X . DML-IV with forests has bias 0.04 and coverage 0.76; what remains is the forests' error in learning h , which shrinks with n as Section 04 discussed. The instrument is an assumption about conditioning as much as an assumption about exclusion, and the conditioning is where machine learning enters.

5.3 The LATE score: apply the AIPW construction twice

With a binary Z and binary D , the conditional LATE of Section 3.5 is a ratio of two average treatment effects of Z : on Y and on D . Each is identified by conditioning on X and has the doubly robust score of Section 04 with the instrument propensity $p(X) = P(Z = 1 | X)$ in the weights (Chernozhukov et al. 2026, Section 13.2):

$$\psi(W; \theta, \eta) = \underbrace{\mu_1(X) - \mu_0(X) + H(p)(Y - \mu_Z(X))}_{\text{effect of } Z \text{ on } Y} - \theta \underbrace{[m_1(X) - m_0(X) + H(p)(D - m_Z(X))]}_{\text{effect of } Z \text{ on } D},$$

with $\mu_z(X) = E[Y | Z = z, X]$, $m_z(X) = E[D | Z = z, X]$, and $H(p) = Z/p - (1 - Z)/(1 - p)$. The score is linear in θ , so $\hat{\theta}$ is the ratio of the two sample means and the influence function is the score divided by the mean of the second bracket, the complier share. Each bracket is doubly robust in the sense of Section 04, and the ratio is Neyman orthogonal (appendix 11.4). This is the doubly robust IV estimator that Section 04's extension list promised.

```
late_dml <- est_dml(dl, "y", "d", c("x1", "x2"), z = "z", model = "iivm",
  learner_mu0 = rf, learner_mu1 = rf, seed = 1, weak_iv = TRUE)
late_dml
#> Double ML estimate (interactive IV model, LATE)
#> Score: AIPW_ratio_LATE; solution: pooled across 5 fold(s); 1 repetition(s)
#> Estimate: 2.1360
#> Std. Error: 0.0701
#> 95% CI: [1.9986, 2.2734]
#> N: 4000 (treated: 1994, control: 2006)
#> Nuisance RMSE: p_hat = 0.487, mu_hat = 1.65, m_hat = 0.395
#> Complier share = 0.609 (SE 0.013); ITT = 1.3002
#> Anderson-Rubin 95% set (bounded): [1.9974, 2.2725]
c(true_late = attr(dl, "late"))
#> true_late
#> 2.045009
```

Table 5: LATE design with compliance and effects that vary with the covariates; $n = 2000$; 100 simulations. Bias is relative to each sample's population LATE.

Estimator	Bias	SD	RMSE	Mean SE	95% coverage
2SLS, linear controls	-0.001	0.133	0.129	0.142	0.990
DML-LATE	-0.002	0.109	0.103	0.103	0.930

In this design the compliance probability and the complier effect both vary with x_1 , and the untreated outcome is quadratic in x_2 . Both estimators are close to the population LATE on average (biases -0.001 and -0.002). The doubly robust score is more precise, with root mean squared error 0.10 against 0.13, because the forests absorb the quadratic term that linear controls leave in the residual. Its standard error also matches its sampling variation (coverage 0.93), while the 2SLS interval over-covers (0.99). The two estimands coincide here because the compliance varies linearly with x_1 ; when it does not, 2SLS targets a linear projection and the score targets the ratio of averages, and only the second is the LATE.

5.4 Observed heterogeneity: the complier CATE

The two brackets of the LATE score are pseudo-outcomes in the sense of Section 08: conditional on X , their means are the conditional effects of Z on Y and on D . Projecting each on a dictionary $p(X)$ and taking the ratio gives the complier CATE $E[Y(1) - Y(0) | X, \text{complier}]$ when it is linear in the dictionary (Chernozhukov et al. 2026, eq. 12.3.1; Abadie 2003). `late_scores()` extracts the two scores and `late_blp()` forms the ratio with delta-method standard errors.

```
ls <- late_scores(late_dml)
late_blp(ls, ~ x1, newdata = data.frame(x1 = c(-1, 0, 1)))[, c("x1", "estimate", "std.error", "conf.low", "conf.high")]
#> x1 estimate std.error conf.low conf.high
#> 1 -1 1.046087 0.10130018 0.8475418 1.244631
```

```
#> 2  0 2.090800 0.06760787 1.9582907 2.223309
#> 3  1 3.135192 0.10401348 2.9313296 3.339055
```

The complier effect in the design is $2+x_1$. The same scores feed `cate_learner()` for a flexible complier CATE and `policy_learn()` for rules whose value is measured among compliers; `grf::instrumental_forest()` is the forest version.

5.5 Proxy controls as a technical instrument

Sometimes the confounder is unobserved but two noisy proxies of it are: two test scores for ability, education and family size for household income. Griliches and Chamberlain’s insight is that one proxy can be put in the outcome equation in place of the confounder, and the other proxy used as an instrument for it, because the measurement errors are independent. In the partially linear form this is `est_dml(model = "pliv")` with the outcome proxy as the endogenous regressor and the treatment proxy as the instrument, and the treatment of interest as an exogenous control (Chernozhukov et al. 2026, Section 12.5). The proximal-inference literature extends the idea to nonlinear models.

5.6 Control function or orthogonal score?

Both start from the same first stage. The control function conditions on $\hat{V}$ inside a parametric second stage and identifies the structural function under an independence assumption on V . The orthogonal score projects on the residualized instrument and identifies an average effect under an exclusion restriction, letting machine learning absorb X . Use the control function when the second stage must be nonlinear and structural (a probit, a demand system, a stock of exposure); use the orthogonal score when the target is an average effect and the controls are many or their functional form unknown. In the linear model with linear controls, they are the same regression.

6 Weak instruments

6.1 Why a small first stage breaks the approximation

With one instrument and controls partialled out, $\hat{\theta} - \theta = \mathbb{E}_n[\tilde{Z}U]/\mathbb{E}_n[\tilde{Z}\tilde{D}]$. The usual approximation treats the denominator as fixed at its limit. When $E[\tilde{Z}\tilde{D}]$ is small relative to the sampling variation of its estimate, the denominator can be near zero, and the ratio is heavy-tailed, biased toward OLS (the numerator and denominator share the endogenous variation), and far from normal. The strength of the first stage is summarized by the concentration parameter $\mu^2 = n\pi^2\text{Var}(\tilde{Z})/\text{Var}(V)$, of which the first-stage F is the sample analogue. Figure 3 shows the sampling distribution of 2SLS for three values.

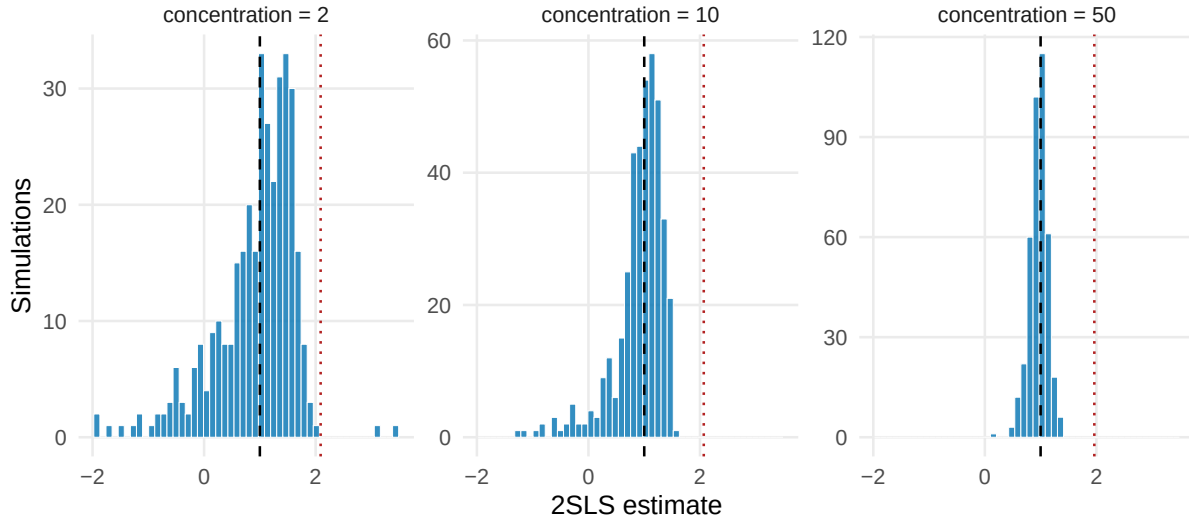


Figure 3: Sampling distribution of the 2SLS estimate (true effect 1, $n = 500$) for concentration parameters 2, 10, and 50. The dotted line is the mean OLS estimate. Estimates beyond three units from the truth are trimmed for display.

Table 6: Weak-instrument Monte Carlo, 400 simulations per design, $n = 500$, true effect 1, nominal 95 percent coverage.

Concentration	Median F	Median 2SLS	Mean OLS	SD 2SLS	Wald coverage	AR coverage	AR set bounded
2	2.744	1.038	2.090	10.106	0.882	0.955	0.355
10	14.257	1.001	2.067	0.493	0.905	0.930	0.953
50	66.584	0.974	1.963	0.163	0.950	0.938	1.000

With a concentration parameter of 2 the median first-stage F is 2.7. The 2SLS distribution is wide and heavy tailed: 23 percent of the samples land more than one unit from the truth, against 0 percent at a concentration parameter of 50, and its mean is pulled toward the OLS value of about 2.1. The Wald interval covers the truth in 88 percent of samples instead of 95. At a concentration parameter of 50 the approximation is fine. The F of 10 in between is the rule of thumb of Staiger and Stock (1997): with F around 10 the bias of 2SLS relative to OLS is about 10 percent. Lee, McCrary, Moreira, and Porter (2022) show that the rule is not enough for inference: across all values of the endogeneity correlation, a first-stage F must exceed about 104.7 for the usual 1.96 critical value to deliver 95 percent coverage, and they tabulate the inflated critical values to use below that.

6.2 Diagnostics and their limits

Report the first-stage F, the robust F, and the effective F of Montiel Olea and Pflueger (2013), which is the statistic to compare with weak-instrument thresholds when errors are heteroskedastic or clustered. `iv_first_stage()` gives all three. Then be careful with what the numbers license. A pretest (“proceed with the Wald interval if $F > 10$ ”) distorts the inference that follows: conditional on passing, the estimator’s distribution is not the unconditional one (Andrews, Stock, and Sun 2019). The diagnostics are for transparency. The decision should be to use inference that does not need them.

```
iv_first_stage(dat, d = "d", z = "z", x = x5)
#> First stage: 1 instrument(s), n = 2000
#> F (homoskedastic) = 3175.61; F (robust) = 2215.85; effective F = 2215.85
#> first-stage t = 47.07
```

6.3 Inference that does not need a strong first stage

The Anderson-Rubin (1949) test never inverts the first stage. For a candidate value θ_0 , form $Y - \theta_0 D$ and test whether the instrument predicts it: under $H_0 : \theta = \theta_0$ the coefficient of $\tilde{Z}$ in a regression of $\tilde{Y} - \theta_0 \tilde{D}$ on $\tilde{Z}$ is zero, and the robust Wald statistic is χ^2 with as many degrees of freedom as instruments, however weak the instrument. Collecting all θ_0 that are not rejected gives a confidence set. With one instrument the set is the solution of a quadratic inequality: bounded when the instrument is strong, unbounded or the whole real line when it is not. That is the honest answer. The reduced-form test ($\theta = 0$ if and only if the reduced-form coefficient is zero) is the special case $\theta_0 = 0$.

```
weak <- sim_iv(500, dgp = "weak", concentration = 3, seed = 6)
ar <- iv_ar_confidence_set(weak, "y", "d", "z", x = paste0("x", 1:3))
ar
#> Anderson-Rubin 95% confidence set (1 instrument(s), n = 500)
#> 2SLS estimate = 0.6987 (SE 1.0367); Wald interval [-1.3332, 2.7306]
#> first-stage F (robust) = 2.17; effective F = 2.17
#> Anderson-Rubin set (unbounded):
#> [-Inf, 1.6432]
#> [6.0884, Inf]
```

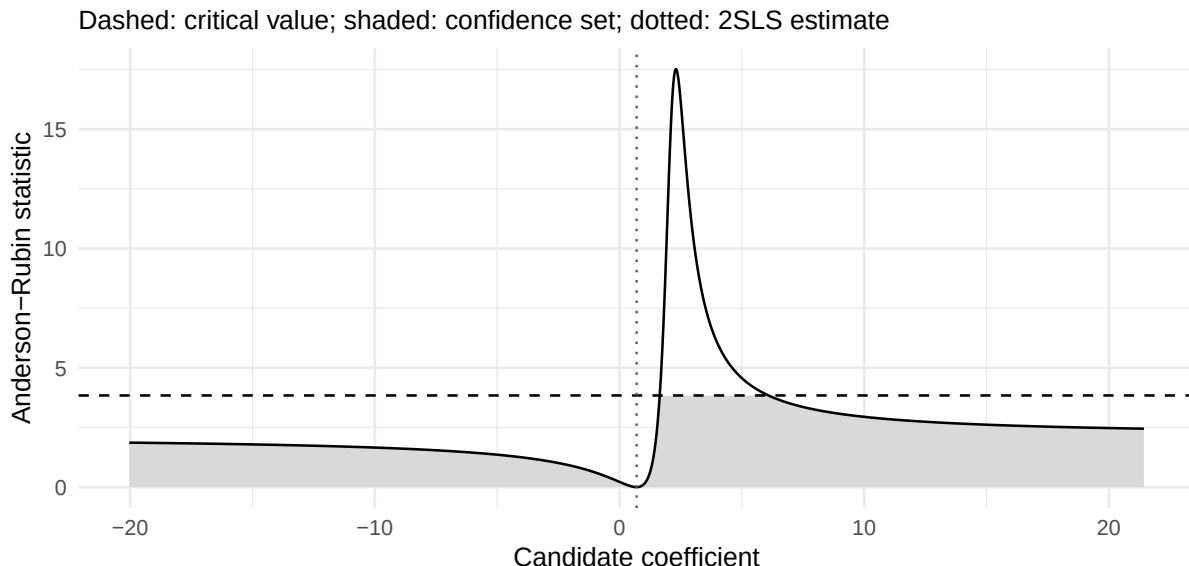


Figure 4: The Anderson-Rubin statistic as a function of the candidate coefficient in a weak-instrument sample. The confidence set is where the statistic lies below the critical value; with a weak instrument it is wider than the Wald interval, asymmetric around the 2SLS estimate, and can be unbounded.

In the Monte Carlo, the Anderson-Rubin set covers the truth in 96, 93, and 94 percent of samples at the three concentration values, at the cost of being unbounded in 64 percent of the weakest samples. Other tests (the conditional likelihood ratio, the Lagrange multiplier) improve power when the model is overidentified; with one instrument the Anderson-Rubin test is optimal (Andrews, Stock, and Sun 2019).

6.4 Weak identification with machine-learned controls

The Anderson-Rubin statistic is Neyman's $C(\alpha)$ statistic for the orthogonal moment: $C(\theta_0) = n \bar{\psi}(\theta_0)^2 / \widehat{\text{Var}}(\psi(\theta_0))$ with ψ evaluated at cross-fitted nuisances. Because the moment is orthogonal, the nuisance estimation error has no first-order effect on $C(\theta_0)$ either, so the test and the set inherit their validity with machine-learned residuals (Chernozhukov et al. 2026, Theorem 13.3.1). The same construction works for any orthogonal score, including the LATE score, which is what `weak_iv = TRUE` computes in `est_dml()`.


```
pliv_rf$weak_iv$set
#>      lower      upper
#> 1 0.9624675 1.081184
c(wald_lower = pliv_rf$conf.low, wald_upper = pliv_rf$conf.high)
#> wald_lower wald_upper
#> 0.9643462 1.0825342
```

With a strong residualized instrument the two intervals agree. The Chernozhukov et al. (2026) example of institutions and growth, with settler mortality as the instrument and a first-stage t of about 2 after forests, is the case where they do not: the robust set's lower end is higher than the Wald interval's, which changes the economic conclusion.

6.5 Exclusion violations are amplified by weak first stages

If the instrument has a small direct effect γ on the outcome, the IV estimand is $\theta + \gamma/\pi$: the violation is divided by the first-stage coefficient. Bound, Jaeger, and Baker (1995) made the point for quarter of birth as an instrument for schooling, where a first stage of 0.1 years multiplies any direct effect by ten. Conley, Hansen, and Rossi (2012) turn it into an interval: allow γ in a range, re-estimate with $Y - \gamma Z$ as the outcome, and take the union of the confidence intervals.

```
pe <- iv_plausibly_exogenous(dat, "y", "d", "z", x5, gamma_grid = seq(-0.2, 0.2, by = 0.1))
pe
#>   gamma estimate std.error conf.low conf.high
#> 1 -0.2 1.430563 0.01701402 1.397217 1.463910
#> 2 -0.1 1.358979 0.01748116 1.324716 1.393241
#> 3 0.0 1.287394 0.01806500 1.251987 1.322801
#> 4 0.1 1.215810 0.01875464 1.179051 1.252568
#> 5 0.2 1.144225 0.01953888 1.105930 1.182521
attr(,"union")
#>      lower      upper
#> 1.10593 1.46391
```

A direct effect of the instrument of 0.2 in either direction, which is a fifth of the first-stage coefficient here, moves the estimate by about 0.07. With a first stage one tenth as large the same violation would move it ten times as much.

7 Many instruments, many effects, constructed instruments

7.1 Several instruments: what 2SLS averages

With several instruments the 2SLS coefficient is a weighted average of the just-identified estimates that use one instrument at a time,

$$\hat{\theta}_{2SLS} = \sum_k \omega_k \hat{\theta}_k, \quad \hat{\theta}_k = \frac{\text{Cov}(Z_k, Y)}{\text{Cov}(Z_k, D)}, \quad \omega_k = \frac{\hat{\pi}_k \text{Cov}(Z_k, D)}{\sum_j \hat{\pi}_j \text{Cov}(Z_j, D)},$$

with $\hat{\pi}$ the first-stage coefficients (appendix 11.7). The weights sum to one but need not be positive. With homogeneous effects every $\hat{\theta}_k$ estimates the same number and the weights are irrelevant. With heterogeneous effects each instrument has its own compliers and its own LATE, and Mogstad, Torgovitsky, and Walters (2021) show that the Imbens-Angrist monotonicity condition applied to the vector of instruments (every unit responds to every instrument in the same direction) is much stronger than it looks. Under the weaker *partial monotonicity* (each instrument is monotone on its own), the 2SLS weights on the instrument-specific LATEs can be negative, and the sign of $\hat{\theta}_{2SLS}$ can differ from the sign of every LATE. Their test checks the sign patterns of the first stages and reduced forms that positive weights require. `iv_late_weights()` reports the decomposition.

```
dat$z2 <- dat$z + rnorm(nrow(dat))
dat$z3 <- rnorm(nrow(dat)) + 0.1 * dat$z
lw <- iv_late_weights(dat, "y", "d", c("z", "z2", "z3"), x = xc)
```

```

lw
#> Two-stage least squares with 3 instruments: estimate = 0.9949 (SE 0.0306), first-stage F = 407.09
#> Instrument-specific IV estimates and 2SLS weights (sum(omega_k beta_k) = 0.9949):
#> instrument first_stage first_stage_t reduced_form cov_zd beta_k omega_k
#>      z      0.99811      24.1015      1.0227 0.9929 0.9966 0.971520
#>      z2     0.02467      0.8619      0.4895 1.0239 0.9524 0.024764
#>      z3     0.02460      0.9028      0.1274 0.1540 0.8369 0.003716

```

Here the effect is homogeneous, so all three instrument-specific estimates are near 1 and the weights only redistribute precision. The instrument with the small first stage gets a small weight; when a weight is negative and the effects differ, the 2SLS number is not an average of anything a policy would deliver. The remedy is to report the instrument-specific estimates and the weights, or to move to the marginal treatment effect framework of Section 7.3, which makes the weighting explicit.

7.2 Judge leniency and examiner designs

The most common instrument in recent applied work is the assigned decision maker: judges for incarceration, examiners for disability insurance, reviewers for patents, loan officers for credit. Cases are assigned at random within a court and year; judges differ in leniency; leniency moves the treatment. The instrument is the leave-one-out leniency of the assigned judge,

$$Z_i = \frac{1}{n_{j(i)} - 1} \sum_{k \neq i: j(k)=j(i)} D_k,$$

computed within the randomization cell and after residualizing D on controls so that leniency is compared among judges who see the same case mix. Leaving out the own case avoids the mechanical correlation between Z_i and D_i that would bias 2SLS toward OLS. Standard errors cluster at the judge level.

```

dj <- sim_iv(6000, dgp = "judge", n_judges = 40, seed = 7)
dj <- leniency_instrument(dj, examiner = "judge", d = "d", x = "x1")
iv_first_stage(dj, d = "d", z = "leniency", x = "x1", cluster = "judge")
#> First stage: 1 instrument(s), n = 6000, clusters = 40
#> F (homoskedastic) = 910.99; F (robust) = 18888.33; effective F = 18888.33
#> first-stage t = 137.43
jfit <- iv_robust(y ~ d + x1 | leniency + x1, data = dj, clusters = judge)
c(estimate = coef(jfit)[["d"]], se = jfit$std.error[["d"]])
#> estimate      se
#> 1.1571504 0.1020343

```

The design's monotonicity condition is the delicate part. Monotonicity across judges requires that a stricter judge incarcerates every defendant a more lenient judge would; judges who are strict on drug cases and lenient on theft violate it. What 2SLS identifies under the weaker *average monotonicity* condition (Frandsen, Lefgren, and Leslie 2023) is a positively weighted average of individual effects only if the judge-specific first stages and effects are positively correlated; the same sign checks as in Section 7.1 apply, and the exclusion restriction (judges affect outcomes only through the decision) is itself testable with the leniency distribution.

7.3 Continuous instruments and the marginal treatment effect

The LATE model with a binary instrument is a threshold model in disguise: monotonicity is equivalent to $D = \mathbb{1}\{p(Z, X) \geq V\}$ with $V \sim U(0, 1)$ independent of (Z, X) , where p is the propensity of treatment and V the unit's resistance to it (Vytlacil 2002). With a continuous instrument the propensity takes many values, and comparing units with propensities p and $p + h$ identifies the effect for units with $V \in (p, p + h]$. Taking $h \rightarrow 0$ gives the marginal treatment effect

$$\text{MTE}(u, x) = E[Y(1) - Y(0) \mid X = x, V = u] = \frac{\partial E[Y \mid X = x, p(Z, X) = u]}{\partial u},$$

the local instrumental variables estimand of Heckman and Vytlacil (2005). This is the control-function approach of Section 4 with the propensity as the control and no functional-form assumption about how the

unobservable enters: $E[Y \mid X, p] = X'\beta_0 + pX'(\beta_1 - \beta_0) + K(p)$ and $K'(u)$ is the part of the MTE that varies with resistance.

Every standard estimand is a weighted integral of the curve (appendix 11.6): the ATE weights all u equally; the ATT weights u by $P(p > u)/E[p]$ (units with low resistance are over-represented among the treated); the ATU by $P(p \leq u)/E[1 - p]$; the LATE of an instrument shift from p_0 to p_1 uniformly on $(p_0, p_1]$; and a policy that changes the propensity from p to p' has the policy-relevant treatment effect with weights $(F_p(u) - F_{p'}(u))/(E[p'] - E[p])$. The curve can be integrated only over the support of the estimated propensity; outside it the estimates are extrapolations of the polynomial.

```
dm <- sim_iv(5000, dgp = "mte", seed = 8)
m <- mte_curve(dm, "y", "d", "z", x = "x1", degree = 3, late_range = c(0.3, 0.7),
              policy = function(p) pmin(p + 0.1, 0.99), n_boot = 99, seed = 1)
m
#> Marginal treatment effects (polynomial, degree 3), n = 5000
#> propensity support: [0.011, 0.971]
#> estimand estimate std.error weight_outside_support conf.low conf.high
#> ATE 1.239 0.1858 0.03518 0.875 1.60
#> ATT 1.690 0.1587 0.02560 1.379 2.00
#> ATU 0.947 0.2515 0.04137 0.454 1.44
#> LATE 1.416 0.0839 0.00000 1.252 1.58
#> PRTE 1.457 0.0556 0.00148 1.348 1.57
c(true_ate = attr(dm, "ate"), true_att = attr(dm, "att"))
#> true_ate true_att
#> 1.249644 1.773894
```

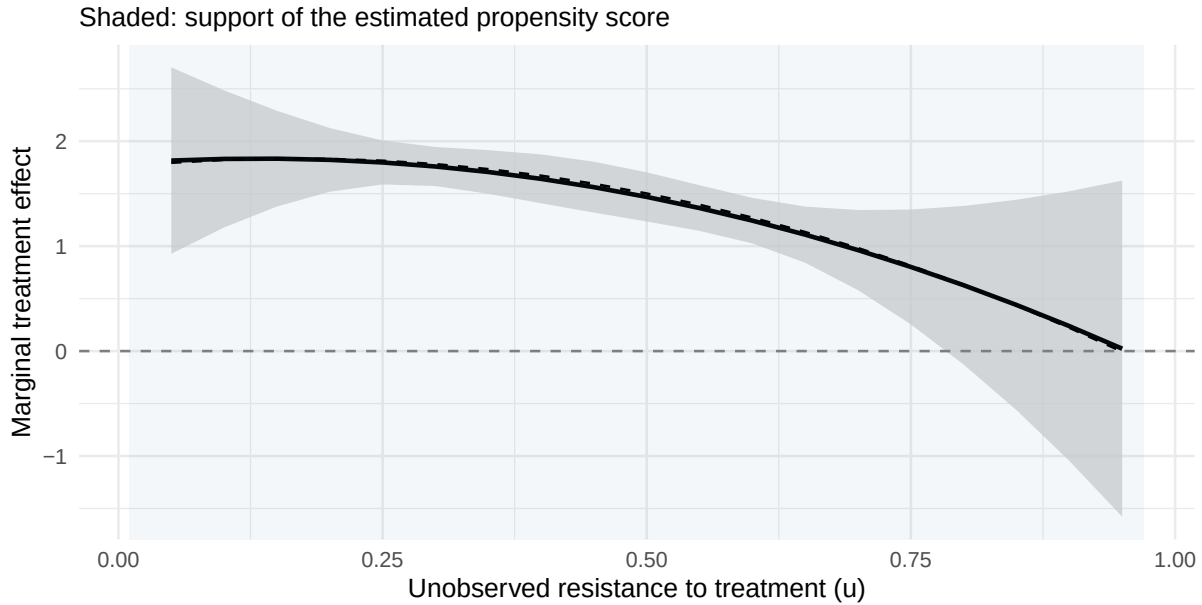


Figure 5: Estimated marginal treatment effect curve (polynomial of degree three in the propensity) with bootstrap bands, and the true curve of the design (dashed). The effect falls with resistance: units easy to move into treatment gain most, which is why the ATT exceeds the ATE.

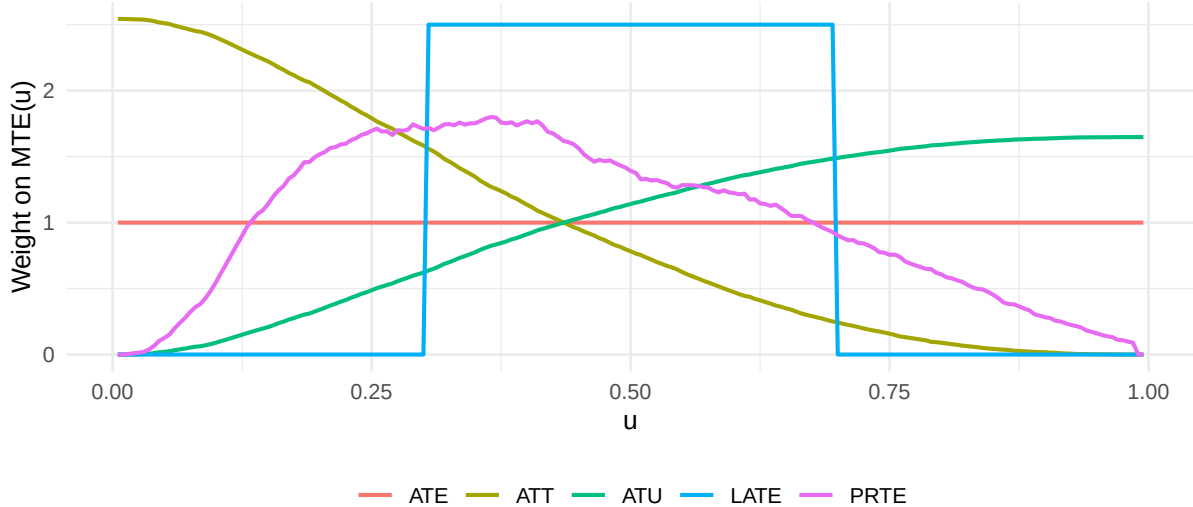


Figure 6: The weights that turn the MTE curve into the ATE, ATT, ATU, the LATE of a propensity shift from 0.3 to 0.7, and the policy-relevant effect of raising every propensity by 0.1.

The MTE framework answers the question the LATE cannot: what would a policy that moves a different margin do? It costs a continuous instrument with wide support and a parametric or smoothing choice for K . Mogstad, Torgovitsky, and Walters (2021) use instrument-specific MTE curves to show that two instruments for college attendance move different people; the replication reproduces that exercise.

7.4 Many candidate instruments

When many variables could serve as instruments, two problems appear. Using all of them in 2SLS adds bias: each instrument adds a little of the endogenous variation back into $\hat{D}$, and with K instruments the bias toward OLS grows with K/μ^2 (Bekker 1994). Limited-information maximum likelihood and jackknife IV (which predicts D_i from a first stage that excludes unit i) reduce it. Choosing among instruments by looking at the outcome invalidates inference.

Belloni, Chen, Chernozhukov, and Hansen (2012) solve both problems with the lasso: select instruments by their first-stage predictive power alone, which is a prediction problem and does not use Y , then run 2SLS with the selected set (or the lasso first-stage prediction as the instrument). Under approximate sparsity the resulting estimator is as good as knowing the relevant instruments. In `est_dml(model = "pliv")` the same idea is the choice of `learner_m` for $E[Z | X]$ and of the instrument set; the `hdm` package implements `rlassoIV()` directly. Gilchrist and Sands (2016) use this to choose among dozens of weather variables as instruments for opening weekend movie attendance, then estimate how much later attendance responds to the exogenous opening shock: social spillovers identified from rain.

7.5 A constructed instrument: shift-share

Many instruments are built rather than found. The shift-share (Bartik) instrument combines local exposure shares with aggregate shocks:

$$B_i = \sum_k s_{ik} g_k,$$

local industry employment shares times national industry growth (Bartik 1991), or, in Autor, Dorn, and Hanson (2013), commuting-zone industry shares times the growth of Chinese imports in other rich countries. Two identification stories justify it and they have different implications.

- **Exogenous shares** (Goldsmith-Pinkham, Sorkin, and Swift 2020): the shares are as good as randomly assigned given controls, and the shocks only provide the weights. Then B_i is a weighted combination of K instruments (the shares), and the estimate is a weighted average of the share-specific just-identified

estimates with Rotemberg weights $\alpha_k = g_k \text{Cov}(s_k, D^\perp) / \sum_j g_j \text{Cov}(s_j, D^\perp)$. The weights say which industries drive the estimate; a few large weights mean the identifying assumption is about those industries.

- **Exogenous shocks** (Borusyak, Hull, and Jaravel 2022): the shocks are as good as randomly assigned across industries, and the shares can be endogenous. Then the estimate equals a shock-level regression of the exposure-weighted outcome on the exposure-weighted treatment instrumented by the shock, and inference should be at the shock level, with as many effective observations as there are independent shocks.

`ssiv_rotemberg()` and `ssiv_shock_level()` compute the two diagnostics.

```
ds <- sim_iv(600, dgp = "shift_share", n_industries = 15, seed = 9)
sh <- paste0("s", 1:15)
rot <- ssiv_rotemberg(ds, "y", "d", shares = sh, shocks = attr(ds, "shocks"), x = "x1")
rot
#> Shift-share IV: 2SLS estimate = 1.0366 (SE 0.0647), n = 600; sum(alpha_k beta_k) = 1.0366
#> share of negative Rotemberg weights = 0.2; sum of negative weights = -0.018; Herfindahl of weights = 0.274
#> top weights:
#> shock alpha_k beta_k
#> s1 0.45943 0.9154
#> s11 0.18707 1.0581
#> s8 0.12961 1.3165
#> s4 0.05663 0.8883
#> s10 0.05226 0.9961
ssiv_shock_level(ds, "y", "d", shares = sh, shocks = attr(ds, "shocks"), x = "x1")
#> Shift-share IV, shock-level regression (15 shocks)
#> estimate = 1.0366 (shock-level SE 0.0754)
#> unit-level 2SLS: 1.0366 (SE 0.0647)
```

The two point estimates coincide by construction; the standard errors do not, because the second counts 600 regions as 15 independent shocks. Which one is right depends on which story the paper tells. The replication of Autor, Dorn, and Hanson (2013) works through both.

8 Instruments in experiments and platforms

8.1 Encouragement designs and A/B tests as instruments

A randomized offer with imperfect take-up is the cleanest instrument there is: independence holds by design, exclusion holds if the offer itself does not change behavior, and monotonicity holds if no one is put off by being offered. The intention-to-treat effect is the reduced form; the LATE is the effect among those the offer moved. Digital platforms run these designs constantly: a recommendation shown at random is an instrument for the action recommended, a nudge is an instrument for the behavior nudged. The tools of Sections 3 and 5.3 apply without change, and the complier profile answers the platform's own question, which is who reacts to the nudge.

8.2 Realized minus expected exposure

Hsieh et al. (2026) estimate the effect of television advertising on purchases from single-source data. Exposure is endogenous: the brand targets shows whose viewers are likely buyers. Their instrument uses the timing of ads within a show. Given that a household watched a portion of a targeted show, whether it saw the ad depends on whether the ad aired during that portion, and the expected exposure is the integral of the ad-timing density over the watched segment. The instrument is realized minus expected exposure, which is mean zero conditional on targeting and viewing, and shifts actual exposure one for one. The second stage is a purchase model with an ad stock, estimated with the control-function stock of Section 4.6. The replication draft simulates the pipeline (exposure, purchase, first stage, second stage) with the public code.

8.3 When instruments are not exogenous shocks

Garcia, Grewal, and coauthors (2022) estimate hotel demand from managerial responses to automated price recommendations. The natural instruments (prices at other hotels) load on correlated demand shocks; the paper argues they are not excluded and instead conditions on the recommendation, which absorbs the manager’s information, using post-double-selection to handle the high-dimensional interactions of recommendations with calendar variables. The lesson for this lecture is that an instrument is a claim about a mechanism, and when the mechanism is not credible the better design is the one that conditions on the information the decision maker used.

9 Traps

9.1 Exclusion failures dressed as natural experiments

Rain is random, but farmers, moviegoers, and voters all respond to rain through more than one channel. Quarter of birth is random, but it is correlated with parental characteristics and with health. The question is never whether Z is random; it is whether Z has *no* other path to Y . Weak instruments make small failures large (Section 6.5).

9.2 LATE is not the ATE

Report the compliers, and say whose effect the number describes. When the policy question is universal (“treat everyone”), the LATE is one point inside the bounds of Section 3.6, and the MTE curve tells whether the marginal units gain more or less than the average.

9.3 Forbidden regressions

Fitted values from a nonlinear first stage plugged into 2SLS, or fitted values from a linear first stage plugged into a nonlinear second stage, are both wrong. The first can be repaired by using the nonlinear fitted value as an *instrument* in a linear 2SLS; the second by the control function (Section 4.3).

9.4 IV larger than OLS

Systematic reviews find that IV estimates exceed OLS far more often than the usual endogeneity stories predict. Measurement error in D explains some (OLS is attenuated); weak instruments and small exclusion failures explain more, because both push IV away from zero in the direction of the reduced-form noise. Young (2022) shows that IV inference is also unusually sensitive to a few high-leverage observations. Report the first stage, the reduced form, a weak-IV-robust interval, and the sensitivity of the estimate to dropping influential units.

9.5 Bad controls, again

Controls measured after the instrument is realized, or affected by the treatment, break the exclusion restriction in the same way they broke conditional exogeneity in Section 03. Judge fixed effects in a leniency design are fine; the judge’s sentence length is not.

10 Practice

10.1 Replications and anchors

- **Mogstad, Torgovitsky, and Walters (2021).** Two instruments for college attendance, the 2SLS weights, instrument-specific MTE curves, and the test of positive weights; `iv_late_weights()` and `mte_curve()`.
- **Hsieh et al. (2026).** Realized-minus-expected exposure as an instrument for advertising, and the control-function stock; `cf_residuals()` with user-written accumulation.

- **Garcia et al. (2022)**. Why external instruments fail for hotel demand and how conditioning on the recommendation with post-double-selection replaces them; `est_dml()` and `hdm`.
- **Gilchrist and Sands (2016)**. Lasso-chosen weather instruments for opening-weekend attendance and social spillovers; `hdm::rlassoIV()`, `iv_first_stage()`, `iv_ar_confidence_set()`.
- **Autor, Dorn, and Hanson (2013)**. The China shock as a shift-share instrument; `ssiv_rottemberg()`, `ssiv_shock_level()`.
- **Oreopoulos (2006)**. The school-leaving age reform as an instrument for schooling, from Wayne’s earlier note; the reporting template of Section 2.5.

10.2 An IV checklist

1. Write the structural equation and say what the error contains. Name the instrument in the abstract and give the mechanism for exclusion.
2. Report the first stage, the reduced form, and the IV estimate with the same controls and sample; cluster where the instrument varies.
3. Report first-stage strength (F, robust F, effective F) and a weak-instrument-robust interval regardless of the F.
4. Say whose effect it is: complier share and characteristics; instrument- specific estimates and 2SLS weights when there are several instruments.
5. With a nonlinear outcome, use a control function and report average partial effects with bootstrap standard errors.
6. With many controls or unknown functional forms, use the orthogonal IV scores with cross-fitting and report the nuisance fit.
7. For a constructed instrument, state which identification story you rely on and report the diagnostics that story implies.
8. Report a sensitivity analysis to exclusion violations, and, if the paper also reports a no-instrument estimate, its sensitivity to unobserved confounding.

11 Appendix

11.1 The LATE theorem and the kappa identities

By consistency $D = ZD(1) + (1 - Z)D(0)$ and $Y = DY(1) + (1 - D)Y(0)$. By independence, $E[Y | Z = 1] = E[Y(D(1))]$ and $E[Y | Z = 0] = E[Y(D(0))]$, so the reduced form is $E[Y(D(1)) - Y(D(0))] = E[(D(1) - D(0))(Y(1) - Y(0))]$ after substituting $Y(d) = Y(0) + d(Y(1) - Y(0))$. Partition by type: $D(1) - D(0)$ equals 1 for compliers, 0 for always- and never-takers, and -1 for defiers, so the reduced form equals $P(\text{complier}) E[Y(1) - Y(0) | \text{complier}] - P(\text{defier}) E[Y(1) - Y(0) | \text{defier}]$. The first stage is $E[D(1) - D(0)] = P(\text{complier}) - P(\text{defier})$. Monotonicity removes the defier terms and the ratio is the complier effect.

For the kappa weights, note that under monotonicity a unit with $D = 1, Z = 0$ is an always-taker and a unit with $D = 0, Z = 1$ a never-taker, and that by independence $E[g(X)D(1 - Z)/(1 - \pi(X))] = E[g(X)D(0)] = E[g(X)\mathbb{1}\{\text{always-taker}\}]$ and $E[g(X)(1 - D)Z/\pi(X)] = E[g(X)(1 - D(1))] = E[g(X)\mathbb{1}\{\text{never-taker}\}]$. Hence $E[\kappa g(X)] = E[g(X)] - E[g(X)\mathbb{1}\{\text{AT}\}] - E[g(X)\mathbb{1}\{\text{NT}\}] = E[g(X)\mathbb{1}\{\text{complier}\}]$, and dividing by $E[\kappa] = P(\text{complier})$ gives the complier mean (Abadie 2003, Theorem 3.1).

11.2 The omitted-confounder bias bound

With $\tilde{A}$ normalized to unit variance, $\gamma = E[\tilde{A}\tilde{D}]$ and $\delta = E[\tilde{A}\tilde{Y}]/E[\tilde{A}^2]$ where bars denote residuals after projecting on $\tilde{D}$. The bias is $\phi = \delta\gamma/E[\tilde{D}^2]$. Writing $R_{D \sim A}^2 = \gamma^2/E[\tilde{D}^2]$ and $R_{Y \sim A|D}^2 = \delta^2 E[\tilde{A}^2]/E[\tilde{Y}^2]$ with $E[\tilde{A}^2] = 1 - R_{D \sim A}^2$ and $E[\tilde{Y}^2] = E[(\tilde{Y} - \beta\tilde{D})^2]$, substitute and simplify to obtain the formula of Section 1.2 (Chernozhukov et al. 2026, Appendix 12.A). The robustness value solves $\phi(r, r) = |\hat{\theta}|$, a quadratic in r .

11.3 Control functions: equivalence with 2SLS and the average structural function

Equivalence. Let $\hat{D} = P_{[Z,X]}D$ and $\hat{V} = D - \hat{D}$. The regression of Y on $(D, X, \hat{V})$ spans the same column space as the regression on $(\hat{D}, X, \hat{V})$, so the coefficient on D equals the coefficient on $\hat{D}$ in the latter. By Frisch-Waugh-Lovell that coefficient is obtained by regressing Y on the residual of $\hat{D}$ after projecting on $(X, \hat{V})$. Since $\hat{D}$ is orthogonal to $\hat{V}$ (a first-stage fitted value is orthogonal to its residual) and X is in the first-stage regressors, the residual of $\hat{D}$ after $(X, \hat{V})$ equals the residual of $\hat{D}$ after X alone, and the coefficient is that of the regression of Y on $\hat{D}$ and X : two-stage least squares. The argument does not use the number of instruments.

Average structural function. If $U = \rho V + \varepsilon$ with $\varepsilon \perp (D, Z, X, V)$ and $\varepsilon \sim N(0, \sigma^2)$, then $P(Y = 1 \mid D, X, V) = \Phi((\alpha + \theta D + X'\beta + \rho V)/\sigma)$, a probit in (D, X, V) with coefficients scaled by $1/\sigma$. The structural probability at (d, x) is $\text{ASF}(d, x) = E_V[\Phi((\alpha + \theta d + x'\beta + \rho V)/\sigma)]$, averaging over the marginal distribution of V ; its derivative in d , averaged over the distribution of (D, X) , is the average partial effect. Replacing V by $\hat{V}$ and the expectation by the sample average gives `cf_ape()`. The generalized residual for a probit first stage is $E[V \mid D, Z, X]$ under the threshold model $D = \mathbb{1}\{z'\gamma + V > 0\}$ with V standard normal, which is $\lambda(z'\gamma)$ for $D = 1$ and $-\lambda(-z'\gamma)$ for $D = 0$, with $\lambda = \phi/\Phi$ the inverse Mills ratio.

11.4 Orthogonality of the IV scores

Partially linear IV. $\psi = (Y - \ell - \theta(D - r))(Z - m)$. The derivative with respect to ℓ in direction h is $-E[h(X)(Z - m_0(X))] = 0$ because $E[Z - m_0 \mid X] = 0$; the same for r . The derivative with respect to m is $-E[(Y - \ell_0 - \theta_0(D - r_0))h(X)] = -E[E[U \mid X]h(X)] = 0$ where U is the structural error with $E[U \mid X] = 0$. The estimator is the ratio of sample means of $\psi_b = (Y - \ell)(Z - m)$ and $\psi_a = (D - r)(Z - m)$, and the influence function is $\psi/E[\psi_a]$; the variance is the IV sandwich on residuals (Chernozhukov et al. 2026, Theorem 13.1.1).

LATE score. Each bracket is the AIPW score of an average treatment effect of Z with the instrument propensity, and Section 04 showed that score is orthogonal to its outcome regressions and its propensity. The ratio $\theta = E[\psi_b]/E[\psi_a]$ is a smooth function of two orthogonal functionals, so it is orthogonal, with influence function $(\psi_b - \theta\psi_a)/E[\psi_a]$ (Chernozhukov et al. 2026, Theorem 13.2.1).

11.5 Anderson-Rubin with cross-fitted residuals

At the true θ_0 the moment $\bar{\psi}(\theta_0) = \mathbb{E}_n[(\tilde{Y} - \theta_0 \tilde{D})\tilde{Z}]$ built from cross-fitted residuals equals its oracle version up to a term whose leading part is a product of nuisance errors (orthogonality) plus a term of order $n^{-1/2}$ times the nuisance error (cross-fitting removes the dependence). Neither involves the first stage, so $\sqrt{n}\bar{\psi}(\theta_0)/\sqrt{\text{Var}} \rightarrow N(0, 1)$ whatever $E[\tilde{D}\tilde{Z}]$ is. Squaring gives the $\chi^2(1)$ limit; with k instruments the quadratic form in the inverse covariance gives $\chi^2(k)$ (Chernozhukov et al. 2026, Theorem 13.3.1). Test inversion over θ_0 yields the set; for $k = 1$ the inequality $n(\bar{m}_Y - \theta\bar{m}_D)^2 \leq c(S_{YY} - 2\theta S_{YD} + \theta^2 S_{DD})$ is quadratic in θ , which is why the set is an interval, its complement, or the whole line.

11.6 Marginal treatment effects and the policy weights

Under $D = \mathbb{1}\{p(Z, X) \geq V\}$ with $V \sim U(0, 1) \perp (Z, X)$ and $(Y(0), Y(1)) \perp Z \mid (X, V)$, $E[Y \mid X, p(Z, X) = p] = E[Y(0) \mid X] + \int_0^p E[Y(1) - Y(0) \mid X, V = u] du$, so the derivative in p is the MTE. For any estimand of the form $E[Y(1) - Y(0)]$ over a population defined by V and p , write it as $\int_0^1 \text{MTE}(u) \omega(u) du$: for the ATE $\omega = 1$; for the ATT, $E[Y(1) - Y(0) \mid D = 1] = E[\int_0^p \text{MTE}(u) du]/E[p]$, so $\omega(u) = P(p > u)/E[p]$; for the ATU, $\omega(u) = P(p \leq u)/E[1 - p]$; for the effect of a policy that moves p to p' , the change in $E[Y] = E[\int_0^p \text{MTE}]$ divided by the change in $E[D] = E[p]$ gives $\omega(u) = (F_p(u) - F_{p'}(u))/(E[p'] - E[p])$ (Heckman and Vytlačil 2005, Table 1). The separable estimator regresses Y on X , pX , and a polynomial in p ; the coefficients on pX are $\beta_1 - \beta_0$ and the derivative of the polynomial is $K'(u)$.

11.7 Weights of 2SLS and Rotemberg weights

With residualized variables, $\hat{D} = \sum_k \hat{\pi}_k Z_k$ and $\hat{\theta}_{2SLS} = \text{Cov}(\hat{D}, Y) / \text{Cov}(\hat{D}, D)$. Expanding the numerator, $\sum_k \hat{\pi}_k \text{Cov}(Z_k, Y) = \sum_k \hat{\pi}_k \text{Cov}(Z_k, D) \hat{\theta}_k$, and dividing by the denominator $\sum_j \hat{\pi}_j \text{Cov}(Z_j, D)$ gives the weights of Section 7.1. For a shift-share instrument $B = \sum_k g_k s_k$, the same expansion with $\hat{\pi}_k = g_k$ gives the Rotemberg weights α_k . The shock-level equivalence of Borusyak, Hull, and Jaravel (2022) follows from writing $\sum_i B_i D_i^\perp = \sum_k g_k \sum_i s_{ik} D_i^\perp = \sum_k s_k g_k \bar{D}_k$ with $s_k = \sum_i s_{ik}$ and $\bar{D}_k$ the exposure-weighted mean, and likewise for the numerator, so the ratio is the shock-level IV coefficient weighted by s_k without an intercept.

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